Planning & Designing Township

How cities and townships work

01
Understanding Cities & Townships

02
What Makes a Great Township?

03
The Evolution of Malaysian Townships

04
The Future of Cities

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The Planning Process/Workflow

06
Key Takeaways & Discussion

Understanding Township Development



The definition of a township varies significantly depending on the places/region.

- In UK, town/township refers to parish division acting as a territorial and administrative unit. A territory and small crofting community.
- In US, town/township functions as a local government division within a county. A specifically measured 36-square-mile area that may or may not align with actual governmental boundaries.

Malaysia's Context (Dasar Perbandaran Negara 2 - DPN2)

- A township is defined as a gazetted area and its adjacent built-up areas, which together must have a minimum population of 10,000 people.
- Additionally, at least 60% of the population aged 15 and older must be engaged in non-agricultural activities. A District Administrative Center, however, still qualifies as a township even if its population falls below the 10,000 threshold.



DPN2 URBAN-CENTRE HIERARCHY

Five levels used to describe the role and influence of urban centres

1	**GLOBAL CITY** BANDAR GLOBAL	Global economic influence • DPN2 threshold: >3.5 million
2	**REGIONAL CENTRE** BANDAR WILAYAH	Influence across a wider region • 1.5–3.5 million
3	**STATE CAPITAL** BANDAR NEGERI	State administration / major state role • no minimum size
4	**MAIN CENTRE** BANDAR UTAMA	Significant urban / economic growth centre • >100,000
5	**LOCAL CENTRE** BANDAR TEMPATAN	Local services • 10,000–100,000; some district centres may be smaller

HOW THIS INFORMS A TOWNSHIP LAYOUT

01 Read the wider role
What city or centre serves this site? What regional jobs, services and transport connections shape its catchment?

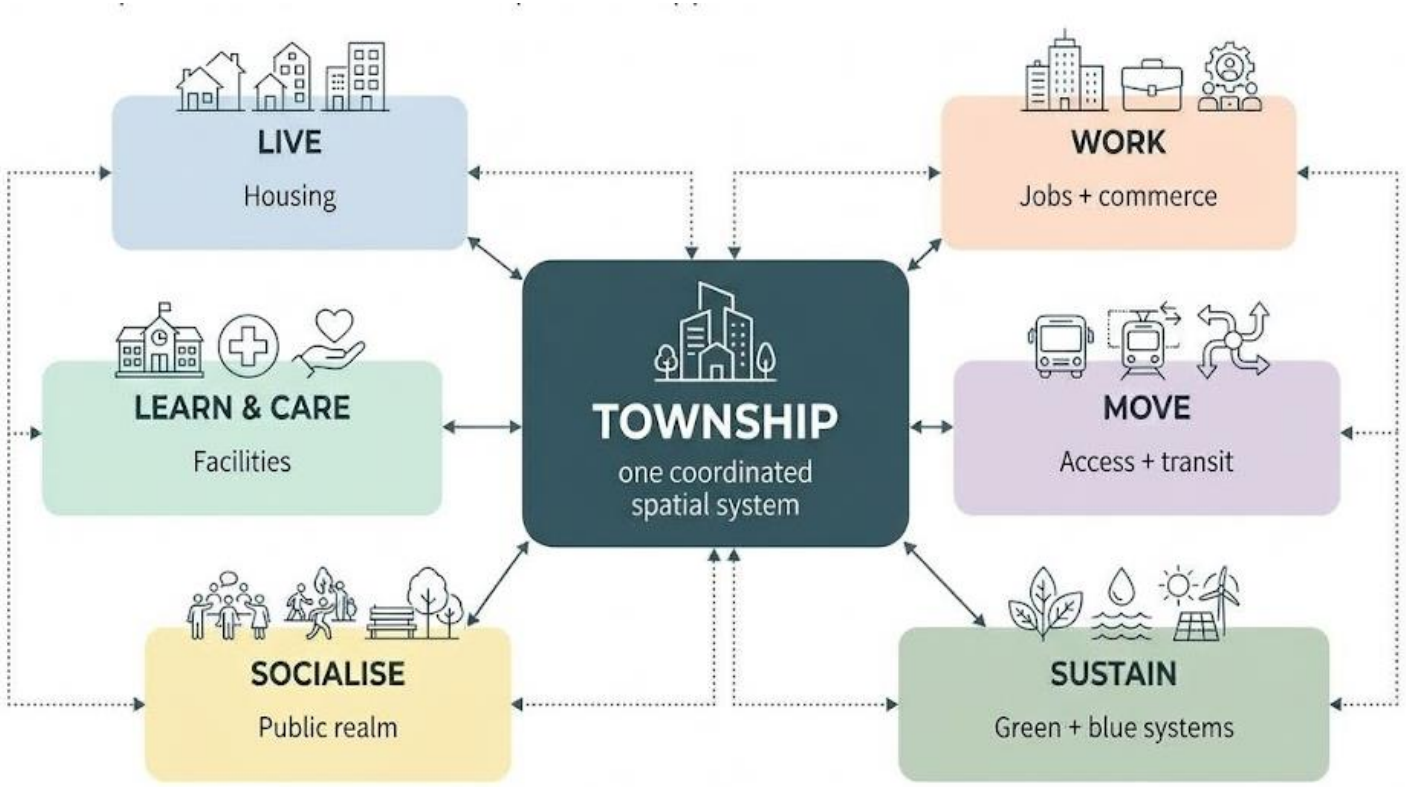
02 Translate role into demand
Estimate the township's likely housing, employment, services, facilities and infrastructure needs.

03 Design the internal structure
Organise town centres, neighbourhoods, movement, green-blue networks and development phases.

IMPORTANT DISTINCTION

"Township" is a development / design scale — not a sixth DPN2 urban-hierarchy tier.

Policy note: DPN2 covers 2016–2025. As of Oct 2026, KPKT lists DPN3 (2026–2035) as under development. Confirm the latest adopted framework for live projects.
Sources: PLANMalaysia, Info Dasar Perbandaran Negara; KPKT, Senarai Dasar KPKT. Population thresholds shown are DPN2 hierarchy criteria, not current population validation.



A township is an integrated urban system

Township development is the planned creation or expansion of an urban settlement through coordinated housing, employment, commercial activities, community facilities, mobility, infrastructure, public spaces and environmental systems.

Think in connected systems - A township is more than the sum of its individual development parcels

What Makes a Successful Township?

Design the layout by the quality of the relationships it creates not by the appearance of the plan alone.

Diversity and connected streets (Jane Jacobs, 1961)
Design for people (Jan Gehl, 2010)



The discussion of a chronicle town development in Malaysia is divided into 3 parts, namely:

British Colonisation & Pre Independence
Melaka
Kuala Lumpur
Taiping

Pioneer
Kuala Kubu
Petaling Jaya Old Town
Bangsar, KL
Subang Jaya
Shah Alam

Interlude
Sunway City
Bandar Utama
Leisure Farm, Johor

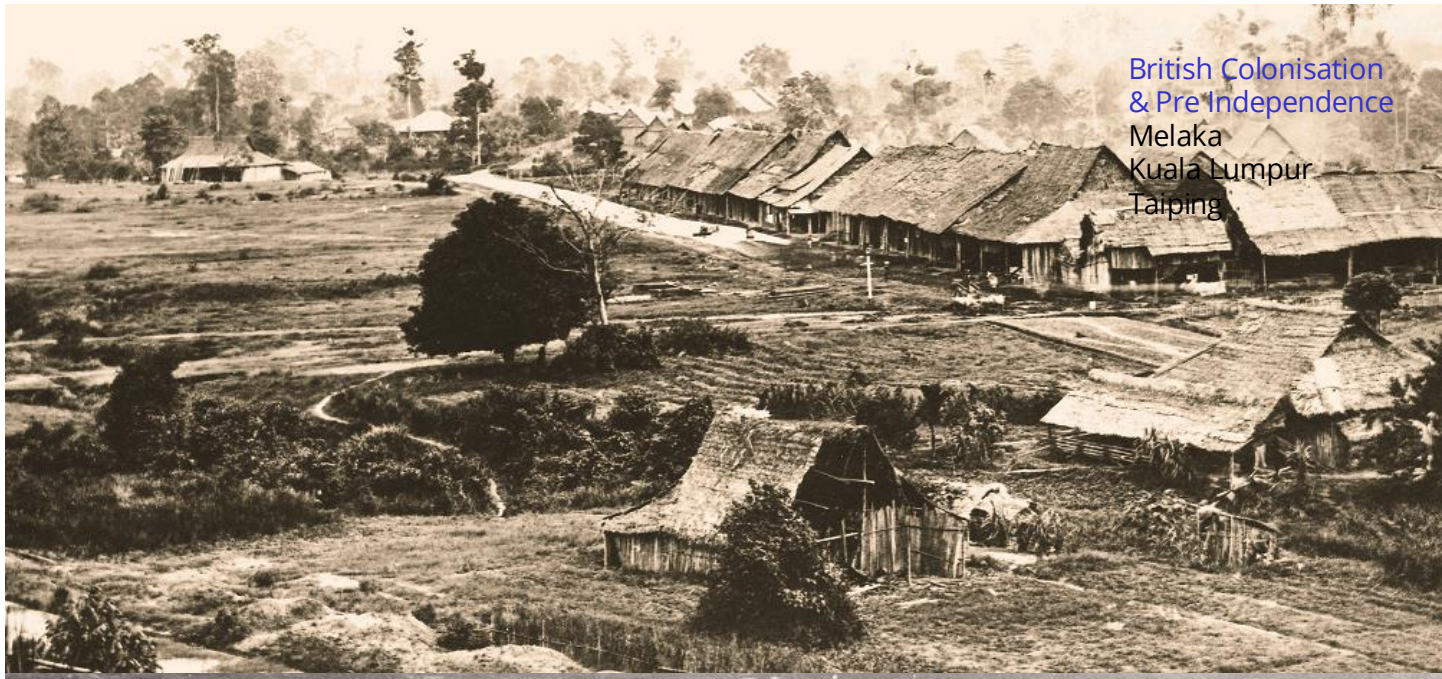
Neoteric
Setia Alam
Bukti Jelutong, Selangor
Alam Impian
Desa Park City
Elmina City

Evolution of Township Development

From Mining Settlements to Urban Centres

Kuala Lumpur and Taiping developed from tin-mining settlements into important urban centres in Malaysia. Their growth was influenced by economic activities, geographical location, transportation networks and administrative functions.

These examples demonstrate how economic opportunities and infrastructure shape urban/town growth, highlighting the importance of integrated planning in guiding the development of modern townships.





Kuala Kubu Bharu



Kuala Kubu Bharu was planned in 1924 by Charles Compton Reade, following repeated flooding in the old settlement, the new town was planned on higher ground, incorporating Garden City principles inspired by Ebenezer Howard.

Its planning approach emphasised a compact town centre, surrounding green spaces, tree-lined streets and a harmonious relationship between the built environment and nature.

Key Planning Principles

- Compact Urban Structure: A clearly defined town centre supported by surrounding neighbourhoods.
- Green Infrastructure: Parks, open spaces and landscaped streets integrated into the town layout.
- Environmental Responsiveness: Site selection and settlement planning influenced by flooding and physical geography.
- Town Identity: A coherent relationship between urban form, greenery and the surrounding landscape.

Shah Alam

Shah Alam was developed to establish a new administrative centre for Selangor. Integrates residential neighbourhoods, industrial areas, commercial activities, civic facilities and a network of major roads.

The Garden City-inspired approach also emphasised green spaces and the relationship between the built environment and nature. As Shah Alam expanded, its planning priorities evolved towards sustainable urban development, environmental quality and improved liveability.

Key Planning Principles

- Planned Urban Structure: Organised neighbourhoods, activity centres and civic functions.
- Economic Integration: Residential development linked to industrial, commercial and employment areas.
- Green City Concept: Parks, landscaped spaces and environmental assets incorporated into urban development.
- Growth Management: Coordinating urban expansion, infrastructure, accessibility and community needs.





Sunway City, Selangor

From Mining Wasteland to Integrated Township, Sunway City is an example of large-scale brownfield regeneration. The development transformed a degraded landscape (abandoned tin-mining) into an integrated township combining residential, commercial, educational, healthcare, retail, hospitality and leisure facilities.

Emphasises mixed-use integration, transport connectivity and environmental rehabilitation. Key initiatives include the Bus Rapid Transit (BRT) system, green spaces and an on-site water treatment plant. The township demonstrates how coordinated land use, infrastructure investment and long-term township management can support an urban environment designed for living, working, learning and recreation.

Setia Alam, Selangor



Setia Alam won the Gold Award in the 2013 FIABCI Prix d'Excellence Awards in the Master Plan category. Setia Alam highlights the importance of considering environmental conditions, accessibility, supporting amenities and residential comfort as part of the overall layout design.

Its planning approach emphasises comprehensive facilities, landscaped green spaces, environmental considerations and residential comfort, creating a self-contained township designed to support diverse lifestyles.

Key Design Principles

- Integrated Township Structure
- Green Spaces and Natural Barriers
- Climate-Responsive Design
- Environmentally Sensitive Infrastructure
- Quality of Life and Community Well-Being

City of Elmina, Selangor

Wellness, Green Living and Nature-Integrated Township Planning

City of Elmina is a master-planned township developed by Sime Darby Property, which integrates residential neighbourhoods, commercial centres, recreational facilities and extensive green spaces within a coordinated urban framework. Its vision combines modern urban living with environmental conservation, community well-being and sustainable development.

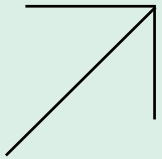
Key Design Principles

- Integrated Township Structure
- Green and Blue Infrastructure
- Wellness-Oriented Planning
- Nature Conservation and Education
- Long-Term Sustainable Development



The evolution of township development/planning in Malaysia reflects changing urban needs and development priorities.

- Early towns were shaped largely by economic activities, geography and administration.
- Planned towns subsequently introduced more deliberate urban layouts, while modern integrated townships combined housing, employment, commercial activities, infrastructure and recreation.
- More recent township concepts emphasis on lifestyle, community interaction, wellness, environmental sustainability and the quality of everyday life.



The challenge today is not simply to develop more land, but to create places where people can live, work, socialise and access everyday needs within a well-integrated urban environment.

Key Takeaway

What will a future-oriented township look like ? ?

Looking ahead, future townships will increasingly combine **people-centred design, climate resilience, green-blue infrastructure and data-informed planning**.

Smart technologies, artificial intelligence and digital systems can support more responsive mobility, efficient resource management, better public services and evidence-based decision-making.

However, technology should remain a means to an end.

The ultimate goal is to **create inclusive, healthy, adaptable and environmentally** responsible communities that **enhance quality of life** while responding to the changing needs of people and future generations.

Is the future township a Smart City ?

Planning & Design Process

The complete planning and design workflow

PHASE 1 — PROJECT INITIATION
Define the brief, objectives, scope, study area and expected deliverables

↓

PHASE 2 — CONTEXT AND FRAMEWORK
Review planning policies, statutory plans, guidelines and surrounding development

↓

PHASE 3 — DATA COLLECTION
Establish the physical, environmental, socio-economic and infrastructure baseline

↓

PHASE 4 — SITE ANALYSIS AND SYNTHESIS
Identify constraints, opportunities, suitability and development capacity

↓

PHASE 5 — DEVELOPMENT PROGRAMMING
Establish population, land-use requirements, densities and facility needs

↓

PHASE 6 — CONCEPTUAL PLANNING
Establish the planning vision, spatial structure, design principles and alternatives

↓

PHASE 7 — MASTER-PLAN DESIGN
Translate the preferred concept into an integrated spatial layout

↓

PHASE 8 — TECHNICAL INTEGRATION
Coordinate transport, drainage, utilities, environment and urban design

↓

PHASE 9 — EVALUATION AND REFINEMENT
Test compliance, feasibility, performance and stakeholder requirements

↓

PHASE 10 — FINAL MASTER PLAN
Finalise the land-use plan, design details, implementation and phasing strategy

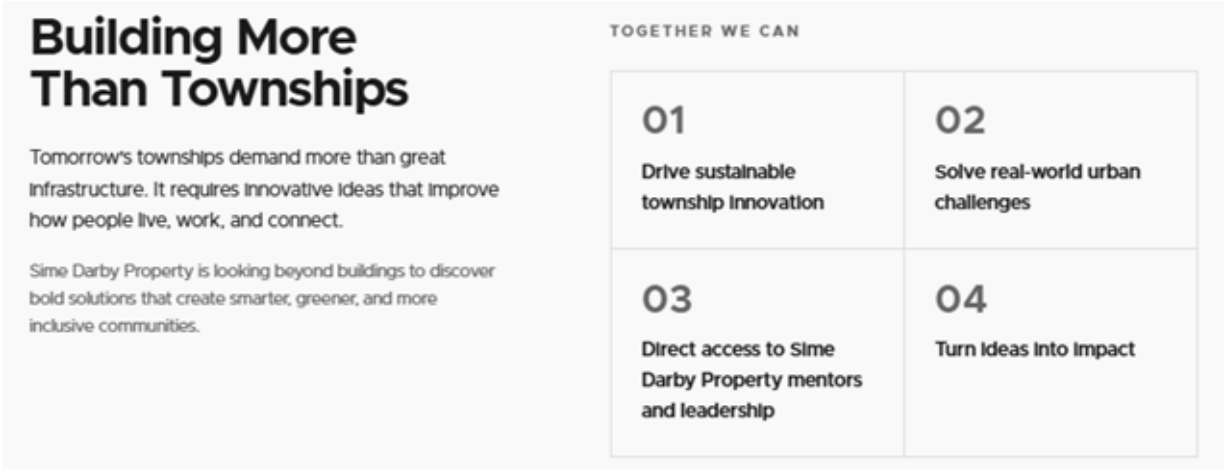
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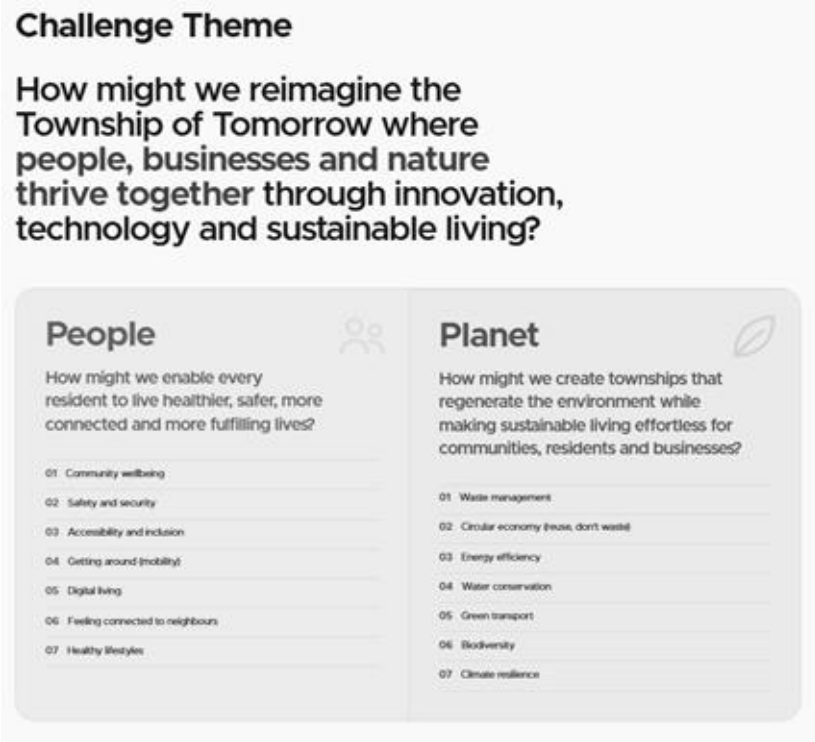
PHASE 11 — DELIVERY AND LEARNING
Prepare the final planning package and establish monitoring and review mechanisms

Phase 1: Project Initiation

Establish what the project must achieve

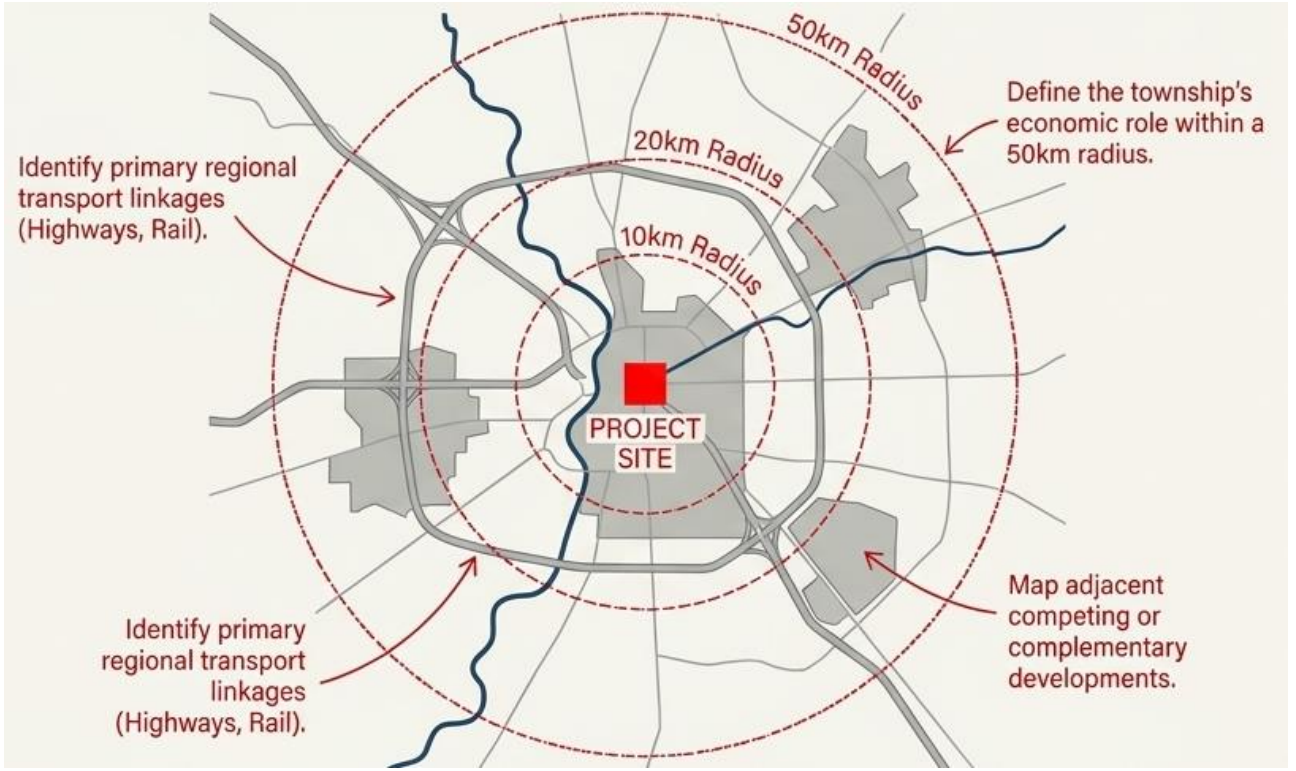
- Identify the **site boundary & profiling**, total land area and **project objectives**.
- Determine the expected population, development type and target users.
- Identify the required **deliverables** such as plans, reports, drawings etc.
- Establish the **project scope & limitations**





Phase 2:

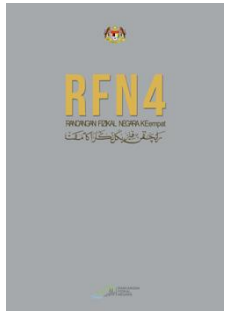
Establish Context & The Planning Framework

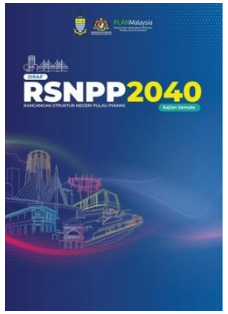


Understand how the development **relates to the local/regional context** and its regulatory context.

- Review related policy and strategic documents: International, National, State, Local etc.
- Review relevant planning standards, guidelines and development controls.
- Identify existing and proposed land-use zoning, accessibility & connectivity networks and surrounding developments.
- Identify land ownership, land status and applicable legal requirements.









Gather reliable information **to understand actual site conditions** through desktop study and fieldwork.

- Physical: topography, soil, hydrology & watercourses, drainage, vegetation etc.
- Accessibility: existing roads, public transport, pedestrian and cycling networks.
- Socio-economic: surrounding population, employment, housing demand, community needs, cultural etc.
- Infrastructure: water, electricity, sewerage, telecommunications and waste management.
- Fieldwork: site observation, survey, photographs, interviews and verification of mapped data.

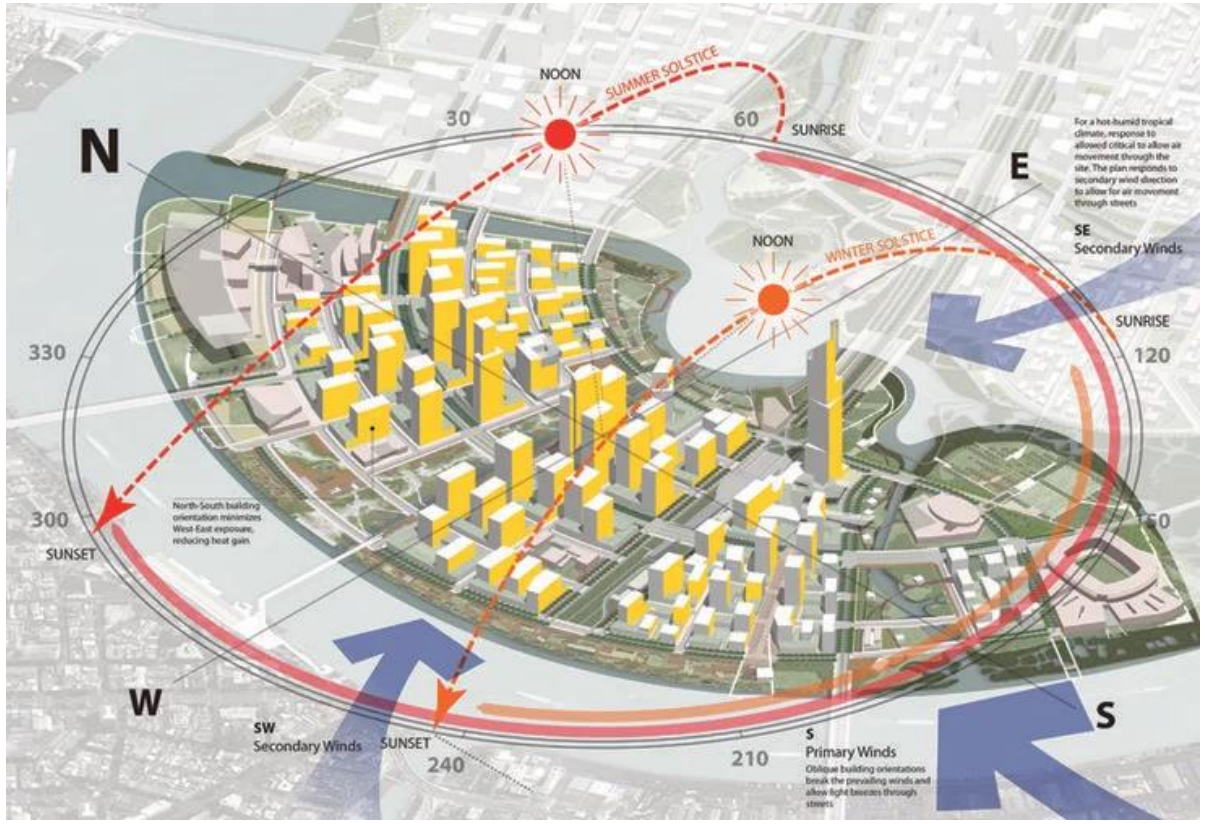
Phase 3:
Data Collection & Site Investigation

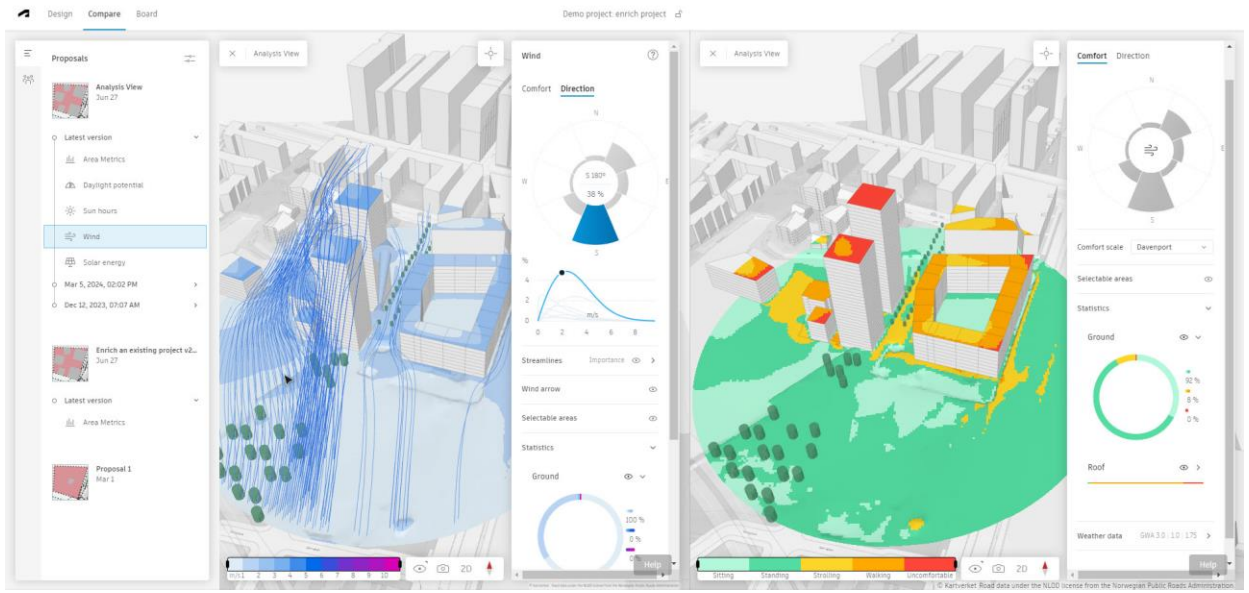


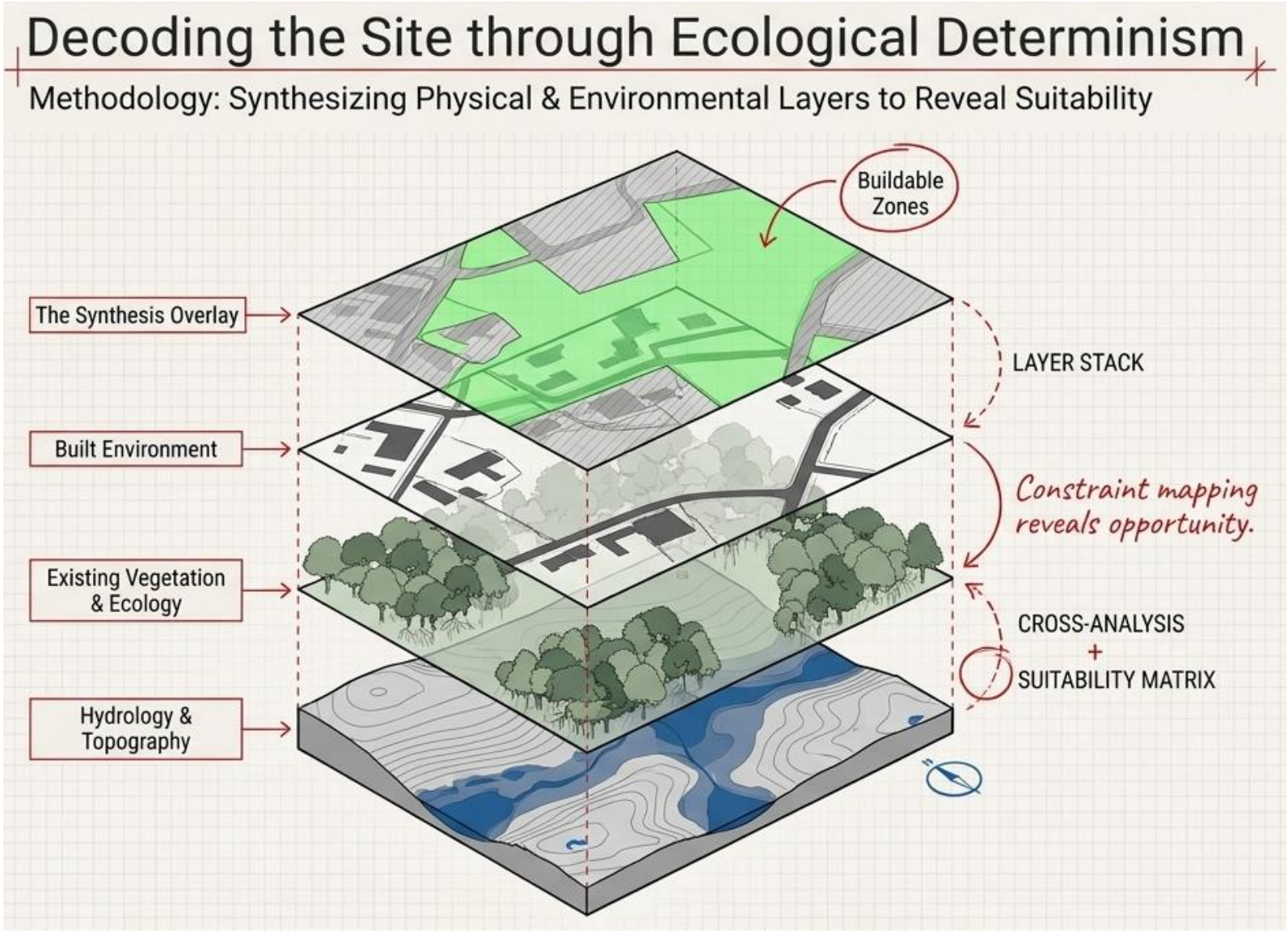
Phase 4:
Site Analysis & Synthesis

Demonstrate how the findings **influence suitability for development, environmental protection and infrastructure provision**

- Thematic analysis: topography, hydrology, ecology, land use, access, utilities, social/economic/cultural context and relevant planning requirements.
- Identify constraints & opportunities, for development
- Relationship with surrounding context (communities, infrastructure etc.)





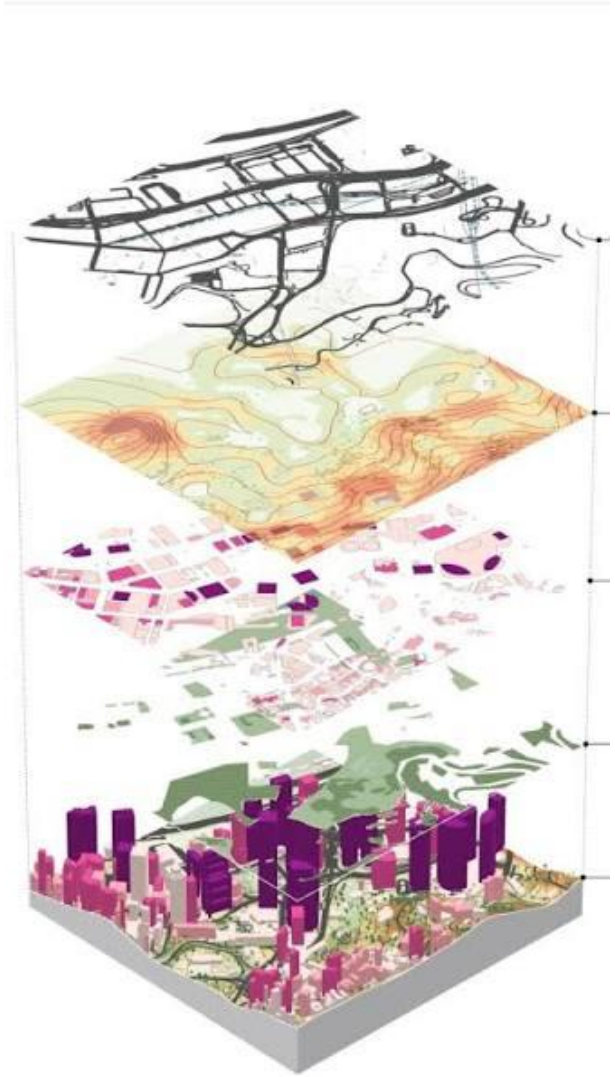


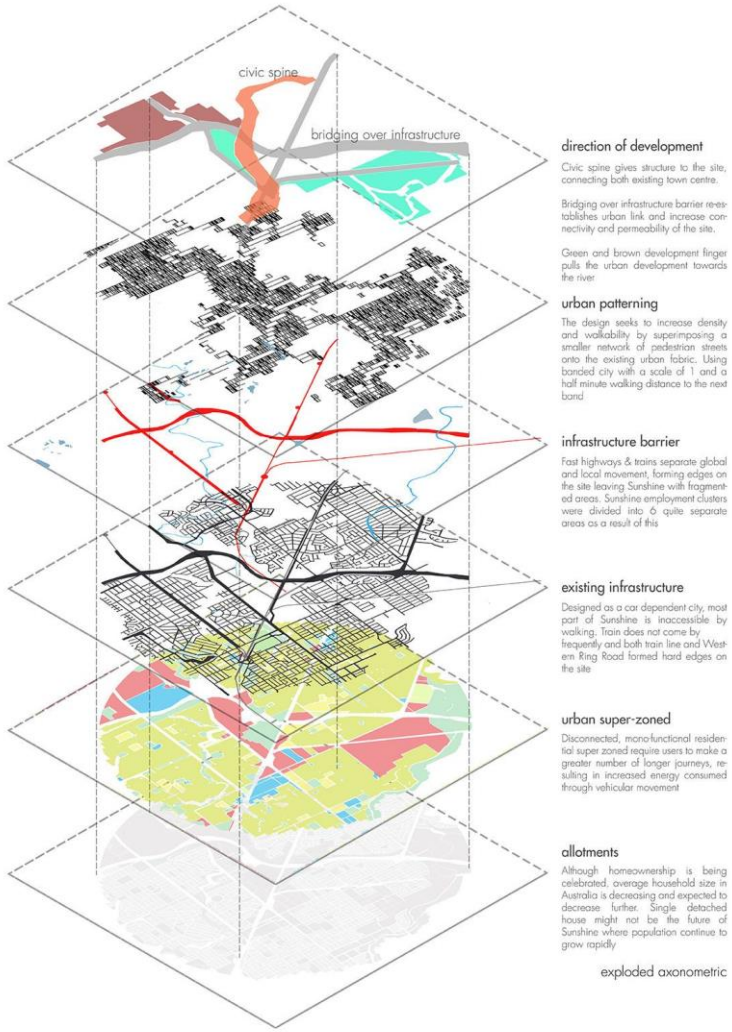
Phase 4:

Site Analysis & Synthesis

- Convert collected data/findings into planning decisions.
- Demonstrate how separate analyses can be synthesised into a constraints and opportunities (Using sieve map).
- Outcome: A site suitability plan (how the analysis informs the proposed spatial structure)







direction of development

Civic spine gives structure to the site, connecting both existing town centre.

Bridging over infrastructure barrier re-establishes urban link and increase connectivity and permeability of the site.

Green and brown development finger pulls the urban development towards the river

urban patterning

The design seeks to increase density and walkability by superimposing a smaller network of pedestrian streets onto the existing urban fabric. Using banded city with a scale of 1 and a half minute walking distance to the next band

infrastructure barrier

Fast highways & trains separate global and local movement, forming edges on the site leaving Sunshine with fragmented areas. Sunshine employment clusters were divided into 6 quite separate areas as a result of this

existing infrastructure

Designed as a car dependent city, most part of Sunshine is inaccessible by walking. Train does not come by frequently and both train line and Western Ring Road formed hard edges on the site

urban super-zoned

Disconnected, mono-functional residential super zoned require users to make a greater number of longer journeys, resulting in increased energy consumed through vehicular movement

allotments

Although homeownership is being celebrated, average household size in Australia is decreasing and expected to decrease further. Single detached house might not be the future of Sunshine where population continue to grow rapidly

exploded axonometric

Phase 5:

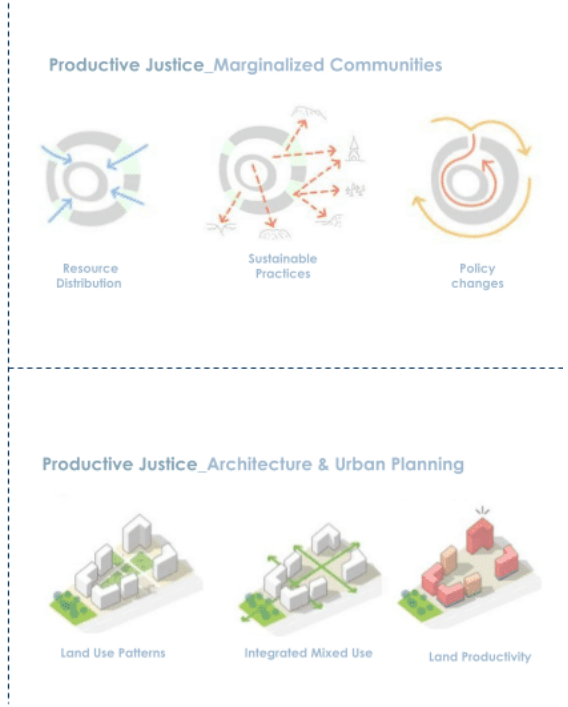
Establish the Development Programme

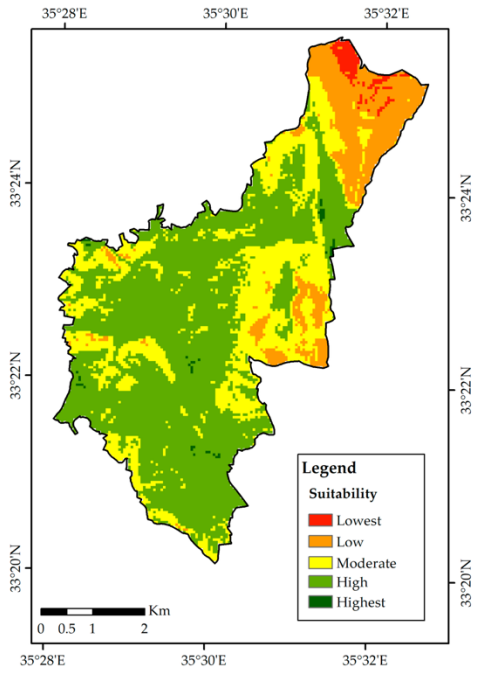
Determine what should be developed, for whom and at what scale.

- Estimate the requirement (based on Planning Guideline & related docs) such as population and housing; dev. intensity and density; and commercial activities; public facilities; open space; and infrastructures/utilities.
- Translate demand into spatial requirements
- The requirements should be derived from appropriate projections, planning standards, technical guidelines and local authority requirements

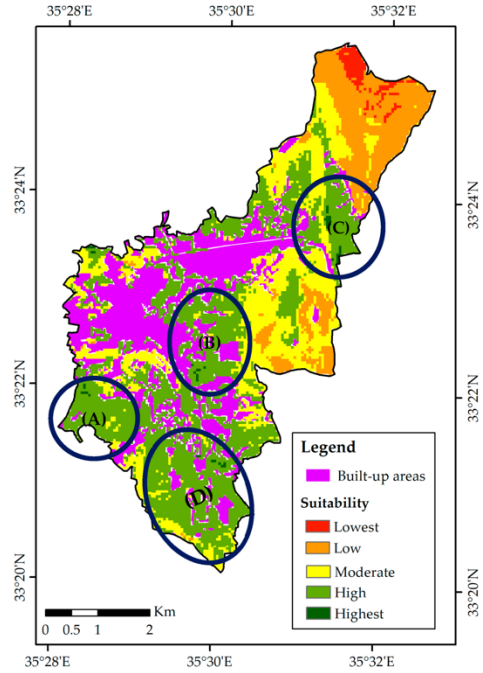



Self Sufficiency
Renewable Energy Sources
Local Food Production
Water Management
Waste Management
AFFORDABLE SOURCE
Focusing on cost-effective renewable options, such as wind or community solar, to ensure energy access for all socioeconomic groups.
ROOFTOP CULTIVATION
Utilizing building rooftops to grow vegetables and herbs maximizes space in dense urban areas and helps reduce heat island effects.
URBAN FARMING
Growing food in city environments using spaces like community gardens or vertical farms to reduce transportation needs and promote local resilience.
LIVE STOCK
Small-scale animal husbandry for meat, dairy, and eggs supports food security and nutrient diversity within communities.
RECYCLE
Reprocessing materials like plastic, glass, and paper minimizes landfill use and conserves natural resources.
ORGANIC WASTE
Managing kitchen and garden waste through composting or digestion helps return nutrients to the soil and lowers methane emissions.
BIODIGESTER
Converts organic waste into biogas and nutrient-rich slurry, offering a clean energy source and reducing landfill pressure.
WATER TREATMENT
Processes like filtration and purification make water safe for consumption and reuse, essential for public health and sustainability.
WATER STORAGE
Storing water in tanks, reservoirs, or underground systems ensures availability during dry periods or supply interruptions.
RAINWATER HARVESTING
Collecting and storing rainwater for non-potable or treated use reduces demand on municipal water systems and supports
SMART GRID
An intelligent power grid that uses digital technology to efficiently distribute electricity and integrate renewable sources.
SOLAR PANELS
Harnessing sunlight through photovoltaic panels or thermal systems provides clean, renewable power with low operational costs.


Productive Justice_Marginalized Communities
Resource Distribution
Sustainable Practices
Policy changes
Productive Justice_Architecture & Urban Planning
Land Use Patterns
Integrated Mixed Use
Land Productivity


35°28'E
35°30'E
35°32'E
33°24'N
33°22'N
33°20'N
Legend
Suitability
Lowest
Low
Moderate
High
Highest
Km
0 0.5 1 2

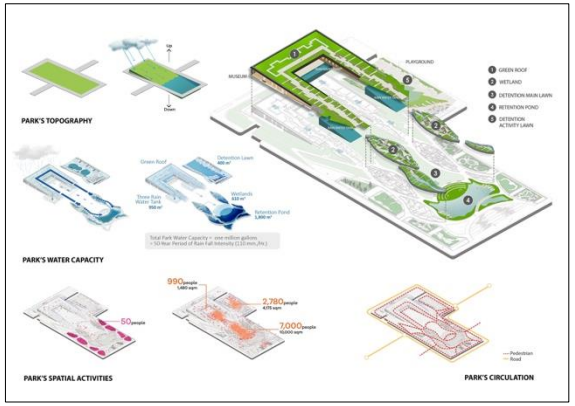
(a)


35°28'E
35°30'E
35°32'E
33°24'N
33°22'N
33°20'N
(A)
(B)
(C)
(D)
Legend
Built-up areas
Suitability
Lowest
Low
Moderate
High
Highest
Km
0 0.5 1 2

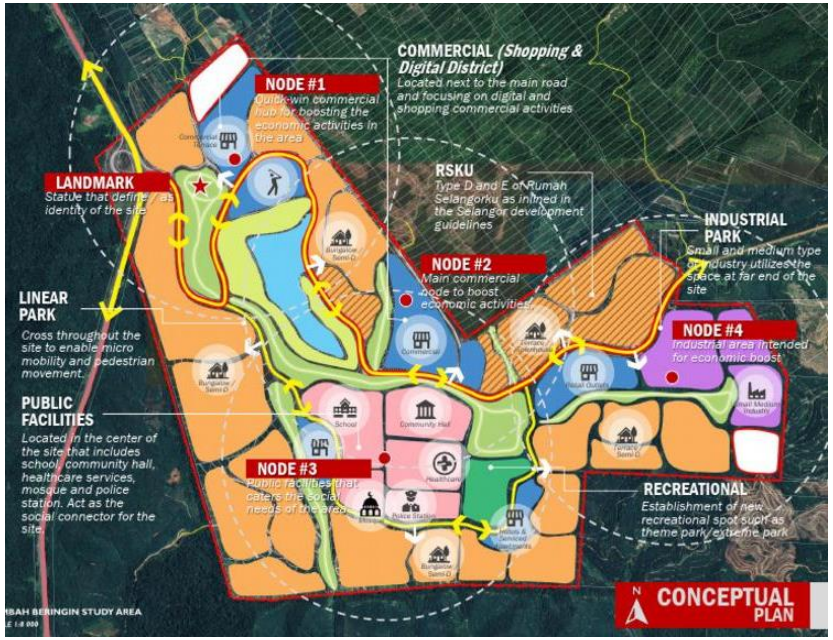
(b)

Phase 6:

Conceptual Planning & Alternative Development







Establish the **spatial logic and overall structure of the development/township**.

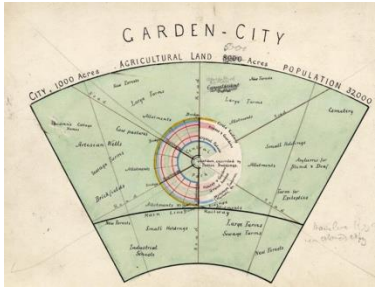
Step 1: Establish the planning vision
To create a connected, mixed-use and environmentally responsive township that integrates neighbourhood living, public transport, accessible public spaces and water-sensitive urban design.

Step 2: Establish design principles that respond to the site analysis and project objectives.

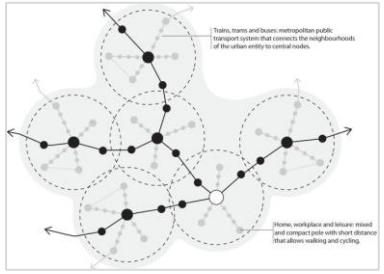
- Mixed-use and active neighbourhoods (Jane Jacobs - urban diversity and the vitality of city streets).
- Human-scale urban design such prioritise pedestrian, active frontages, shade, seating and attractive public spaces. (Jan Gehl - designing cities for people).
- Connectivity and transit-oriented development
- Green-blue infrastructure such as integrate parks, water-sensitive urban design.

Step 3: Explore and **develop alternative spatial concepts.**

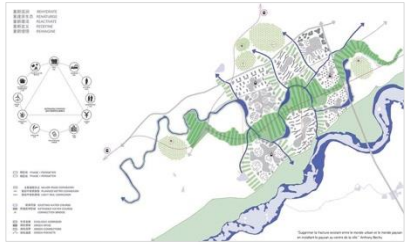
Option A - Centralised development
Strengths: Strong central identity and concentrated services.
Limitations: May create a single dominant centre and concentrated traffic



Option B - Polycentric development
Strengths: Distributed access to services and opportunities for phased development.
Limitations: Requires careful coordination of centre hierarchy and transport connections.



Option C - Landscape-led development
Strengths: Support ecological connectivity, landscape character
Limitations: Requires careful testing of developable land, connectivity and infrastructure costs.



Step 4: Evaluate and select the preferred concept. Use an evaluation matrix to **compare alternatives** against agreed criteria.

Evaluation criterion

- Planning compliance
- Accessibility
- Land-use efficiency
- Environmental performance
- Urban design
- Infrastructure feasibility
- Economic feasibility
- Flexibility

“Which option best meets the objectives and what evidence supports the decision?”

Phase 7:

Masterplan Design



The **core design phase** which **translates the selected concept into a coherent development layout**. It is where the planner coordinates across several interconnected design components.

- Land-use and urban form: Establish compatible land-use (residential, mixed-use, commercial etc) relationships, development intensity and transition zones.
- Movement and connectivity: Establish road hierarchy, street network, public transport connections and facilities.
- Neighbourhood structure and facilities: Consider catchment areas, walking access, demand and the needs of different users.
- Public realm and landscape: Integrate green-blue infrastructure that are accessible, connected, usable and appropriately located.
- Development parcels & zones: Establish parcel dimensions, plot subdivisions, building orientation and interfaces with streets and public spaces.

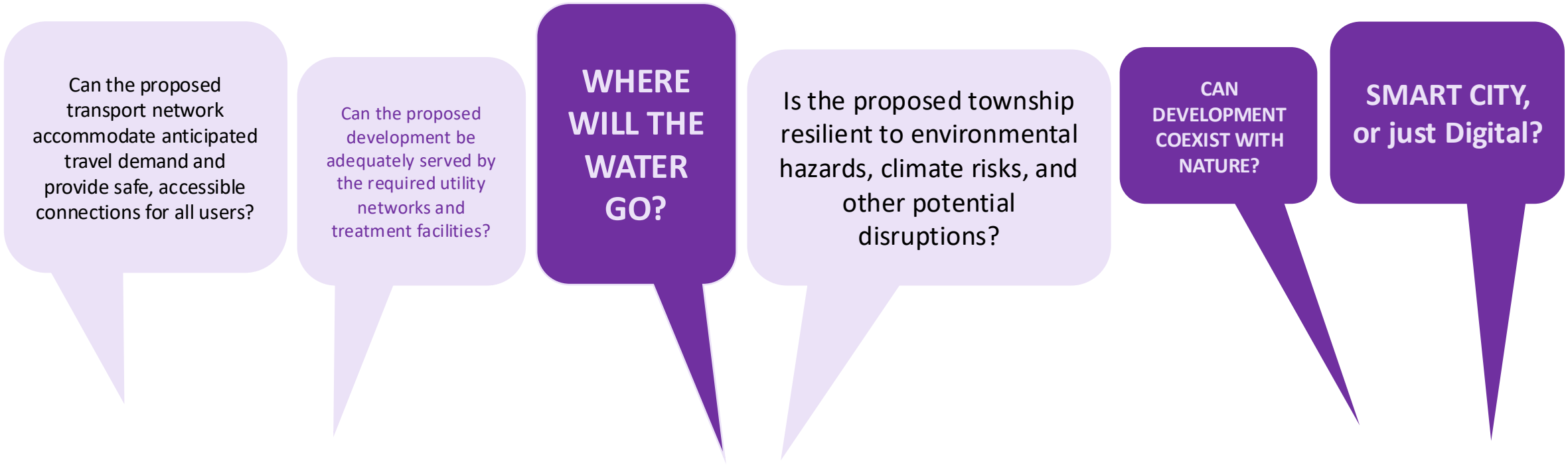
These components must be developed together rather than sequentially in isolation.

Phase 8:

Masterplan Design

To ensure that the master plan functions as a coordinated physical system, the design process is **non-linear and iterative**, involving **continuous cycles of review, evaluation, and refinement** rather than following a linear sequence.

Design components are interdependent: a change in one can affect several others.



Phase 9:

Evaluate & Refine the Masterplan

Review the consistency of land use, movement, public spaces, facilities, drainage, utilities, development intensity and phasing. Revisit earlier decisions when a conflict is identified; do not assume that supporting plans will automatically fit together.

A strong plan must be attractive, compliant, feasible and capable of delivering its objectives.

Compliance

- Check land-use and development controls
- Verify density, plot ratio and reserves
- Confirm access and facility requirements

Performance

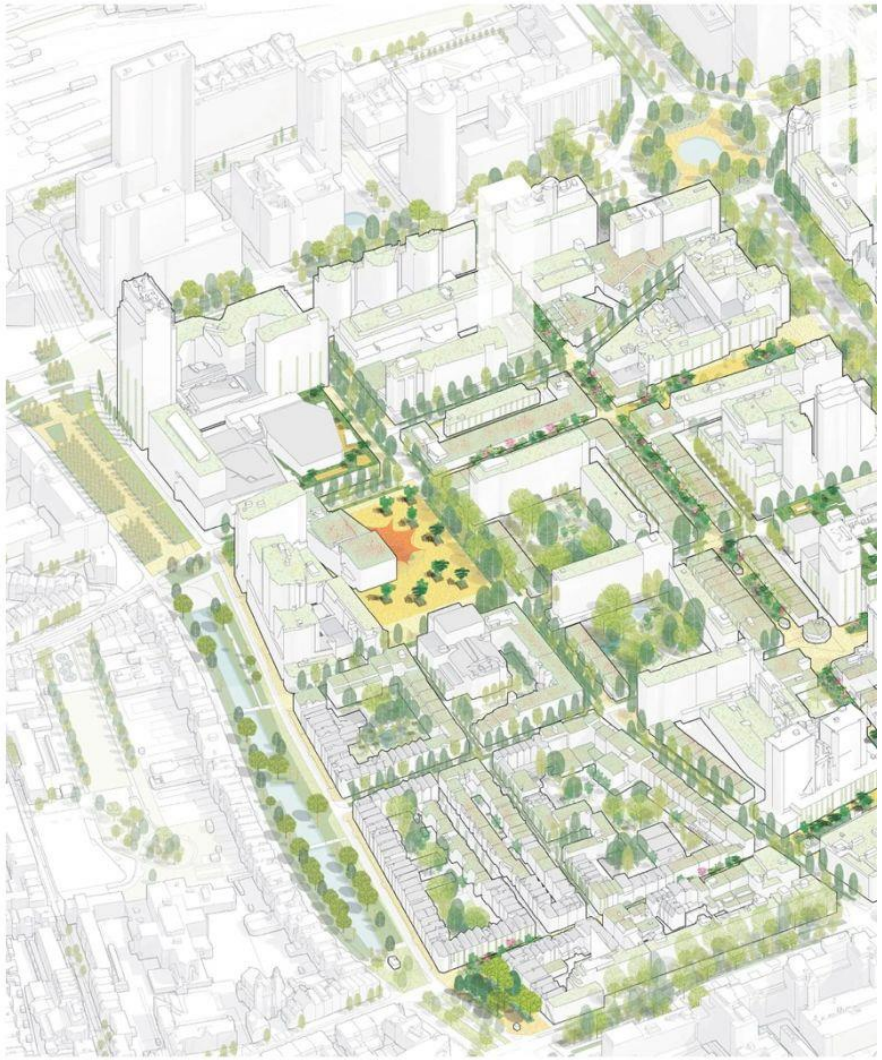
- Evaluate accessibility and connectivity
- Review public-space distribution
- Assess environmental outcomes/impacts

Refinement

- Resolve conflicts and gaps
- Revise affected parcels or networks
- Re-evaluate against objectives

Phase 10:

Phasing & Final Masterplan Delivery



Phasing Strategy

Determine the logical order of development based on:

- Access and infrastructure readiness.
- Development demand and financial feasibility.
- Dependencies between infrastructure and development parcels.
- Early delivery of essential facilities.
- Environmental protection and construction sequencing.

Delivery & Monitoring

The planning and design workflow should consider how the completed development will perform, not simply whether the drawings have been completed.

- Compliance to required statutory submissions and approval processes.
- Coordinate with relevant professionals.
- Monitor construction and compliance with approved plans and conditions.
- Evaluate actual performance against planning objectives.
- Identify lessons and implement adjustments where appropriate.

A strong plan (township) is reasoned, coordinated and deliverable.

01	**Understand before drawing**	Start with the brief, framework and evidence.
02	**Design relationships**	Coordinate land use, movement, public realm and infrastructure.
03	**Evaluate and revise**	Use technical findings and evaluation to improve the plan.
04	**Plan for delivery**	Connect the proposal to phasing, responsibilities and monitoring.

Zulkifli Ahmad Zaki
(Town & Regional Planing)
Centre of Studies for Architecture & Planning
Faculty of Built Environment / Fakulti Alam Bina
UiTM
zulkifliahmadzaki@uitm.edu.my